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ENGINEERING



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Saveetha Institute of Medical And Technical Sciences
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OFDMA Bandwidth & Latency Trade-offs

CO3 AT2 TECHNICAL PRESENTATION

Question:

"Derive the total OFDMA bandwidth for 1200 subcarriers at 15 kHz spacing, then repeat the derivation for 30 kHz spacing, showing both calculations clearly. Compare the two resulting bandwidths and explain the latency-vs-bandwidth trade-off: wider spacing shortens symbol duration (lower latency) but consumes more spectrum for the same subcarrier count. Discuss which use case (eMBB vs. URLLC) favors each option."

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OFDMA Bandwidth Derivation

The total bandwidth required is the product of the number of subcarriers and the subcarrier spacing.

$$\text{Bandwidth} = N_{\text{subcarriers}} \times \Delta f$$

15 kHz Spacing

Number of Subcarriers = 1200

Spacing = 15 kHz

$$BW_{15} = 1200 \times 15 \text{ kHz} \quad \mathbf{BW_{15} = 18,000 \text{ kHz} = 18 \text{ MHz}}$$

30 kHz Spacing

Number of Subcarriers = 1200

Spacing = 30 kHz

$$BW_{30} = 1200 \times 30 \text{ kHz} \quad \mathbf{BW_{30} = 36,000 \text{ kHz} = 36 \text{ MHz}}$$

Comparison: Doubling the subcarrier spacing exactly doubles the required bandwidth.

Latency Trade-offs & 5G Use Cases

The Inverse Relationship

Symbol duration is inversely proportional to subcarrier spacing.

$$T_s \propto \frac{1}{\Delta f}$$



The Trade-off: Wider spacing (30 kHz) shortens symbol transmission time, resulting in lower latency, but requires twice the spectrum (36 MHz vs 18 MHz).



eMBB (Enhanced Mobile Broadband): Favors 15 kHz.

It maximizes spectral efficiency for high data rates where extreme low latency is not the primary constraint.

URLLC (Ultra-Reliable Low-Latency): Favors 30 kHz (or



higher). Rapid symbol transmission minimizes delay, crucial for autonomous driving or industrial automation.

